**Eigen**Engine

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# Persistence and generation as problems of representation

EigenEngine, with Gerard McCaul

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Generated by the engine it describes.

*A language model forgets between sessions and, left to itself, writes toward the average of its training. We describe a system that pays the cost of organisation once, when documents enter, by folding them into a graph of concepts organised into levels of increasing generality (a graph whose nodes are labelled by level of generality, from raw facts up to broad themes), whose small top layer answers structural questions by arithmetic alone, with no model run at query time. On a corpus of 1,307 papers the structure compresses geometrically, the number of levels a query must descend stays near five regardless of corpus size, and a single query recovers thematic connections between researchers – cases where two scientists' work clusters into the same concept node – most of which are invisible to citation search because neither has cited the other, because thematic overlap is read off as a set intersection over pre-built summary nodes and no new document is needed to hold the link. The persistence problem is solved by this system; the generation problem – producing prose in a personal voice – is not. Section 5 marks the boundary. This document was drafted by the system it describes.*

## 1 The two gaps

Two structural gaps limit what a language model can do for a knowledge worker who returns to the same body of work across many sessions.

The first gap is persistence. A language model has no state that survives the end of a session. Each new session receives only what is placed in its context window, the fixed-length input it processes before generating a reply. To recover prior work, the entire relevant history must be re-supplied. Context windows are bounded, so complete re-supply is eventually impossible, and on long inputs recall degrades toward the middle of the window – language models attend less reliably to content placed there – the effect documented in retrieval benchmarks as *lost in the middle* [1]. The model cannot accumulate a structured record of a body of work that grows over time.

The second gap is generation. A language model keeps no record of what it has previously asserted, so a new output cannot be checked against its past for contradiction. Lacking that check, it falls back on its training and returns whatever continuation the text it was trained on makes most likely, rather than something forced on it by evidence it did not already hold. Its output drifts toward the average of everything it has read [2].

Steering a fixed model does not close either gap. Prompting changes what the context window holds but not whether it survives the session. Fine-tuning moves the weights globally and cannot hold one specific, growing body of work; trained further on new material, a network overwrites what it knew, the catastrophic-forgetting failure named in connectionist systems decades ago [3–5]. Reinforcement optimises toward a reward [6], but in the closed-loop setting where the reward model is trained on human preferences over model outputs, the reward signal ranks continuations the model already produces and introduces no external grounded evidence [7]. Generation requires evidence the system does not yet hold. The same model will take a new task from a few examples in its context and never move a weight [8, 9]. Each intervention leaves the store absent.

## 2 Representation

The same meaning can be expressed in two forms, as natural language or as a vector in a linear space, and a question that is computationally expensive in one form can become a single arithmetic operation in the other. A message crosses the gap between two minds only as far as the code they share will carry it [10], and the representation worth having keeps only what predicts the target, which written down is the information bottleneck [11, 12]. Let $T$ be a compressed representation of the input $X$, produced by the stochastic encoder $p(t \mid x)$; the target variable to be predicted is $Y$. Mutual information $I(X;T)$ measures how much knowing $X$ tells you about $T$ (in bits); $\beta$ is a dial – large $\beta$ demands

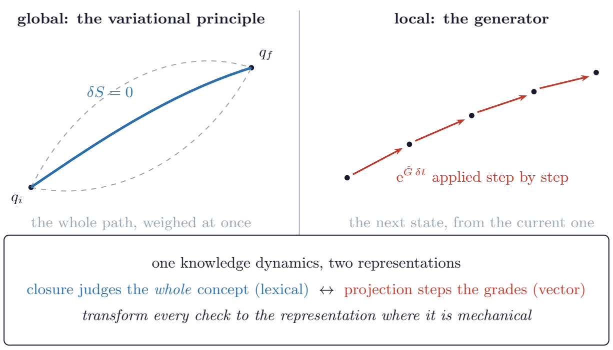


**Figure 1:** One knowledge dynamics, two representations. The left panel shows the variational principle: a single extremal path ($\delta S = 0$) selected from a family of varied paths. The right panel shows the generator $e^{\hat{G}\delta t}$ advancing one step at a time from the current state. Closure checks go left; projection steps go right; each check is sent to the representation where it is arithmetic. (*Closure*: a check that every concept used in an output has been defined before use.)

fidelity, small $\beta$ demands compression. The bottleneck then writes as

$$\mathcal{L}[p(t \mid x)] = I(X;T) - \beta\, I(T;Y), \tag{1}$$

the code $T$ that forgets as much of the input $X$ as it can while keeping what it must about the target $Y$.

Containment – whether one concept encompasses another – requires argument in prose; in a vector space it reduces to a single vector comparison [13], where a concept's vector is componentwise no larger than the vector of any concept it contains. Whether two records refer to the same person requires an essay in words; in a set representation it reduces to a membership test [14]. The structure built in Section 4 is organised so that these are the operations actually used at query time.

The structure is built directly: a graph whose nodes are concepts and whose edges record which concepts contain which, which appear together, and who wrote what, stored so that a query is an arithmetic lookup over pre-built structure. Documents are grouped from the bottom up into broader and broader themes, so the top layer holds a small number of summary nodes, one per theme, each carrying only the features its members share. A query answerable at that top layer never descends to the documents beneath it.

## 3 Growing the representation backwards

The graph is built in the opposite direction from standard machine-learning practice. In the forward direction, data is supplied first and a function is fitted to it; structure, if it is recovered at all, must be read back out of the fitted weights afterwards. The approach here inverts that order. Before a document is stored it is assigned to a cluster, and before a cluster is stored it is assigned to a theme. The cost of deciding where a piece of information belongs is paid once, at ingestion. Once assigned, retrieval is arithmetic and the query traverses structure.

Distillation trains a small model to carry a large one's function rather than re-earn it [15]. A deep network is a tower of coarse-grainings – the same move that in physics asks which microscopic details survive averaging into macroscopic laws [16, 17]. And the brain does not store memory: it rebuilds it each time out of structure [18, 19], running a fast store beside a slow one that folds the day into the shape of the whole, because any single network made to swallow everything at once forgets the lot [20, 21].

## 4 The graded structure and the cost of recall

The folding operation assigns each document's atomic propositions to a cluster, and each cluster to a theme. We call the items assigned to a node its children; the node is their parent. The resulting graded graph – the *conceptric*, a persistent structure encoding one body of knowledge in levels of generality – is built by repeating this move. At the lowest level sit the elementary claims extracted from each source document at ingest time – sentences or short passages recording a single fact or assertion (hereafter atomic propositions). Propositions are grouped into clusters, clusters into themes; at each level the members of a group are replaced, for upward queries, by a single summary node encoding what they share (Figure 2). The shape biology reports for semantic memory is the same: to recall a fact is to descend from the general to the particular [22, 23]. Both shapes may follow from the same constraint – what survives when a world is forced through a finite description – though a full derivation remains in preparation [24].

Recall is then a descent, the recursive shape the retrieval literature has approached from the opposite direction [25–27]. Earlier hierarchical retrieval methods (e.g., RAPTOR [26]) cluster passage embeddings but rebuild the index at each query; here the hierarchy is built once at ingest and structural queries cost zero model calls. The grading is forced by arithmetic: a flat index over $N$ items searches in time growing with $N$ unless the index is itself hierarchical [28]. If each node at one level covers $b$ children at the level below, then $k$ levels cover $b^k$ items; setting $b^k = N$ and inverting gives $k = \log_b N$. A graded hierarchy of branching factor $b$ therefore reaches any item in

$$d = \lceil \log_b N \rceil \tag{2}$$

steps. Structural queries, such as which research themes two scientists share, are answered by set operations over the theme-summary nodes alone; the document-level records are never read and no language model is invoked. If Author A's theme-summary node contains clusters $C_1$ and $C_2$, and Author B's contains $C_2$ and $C_3$, they share

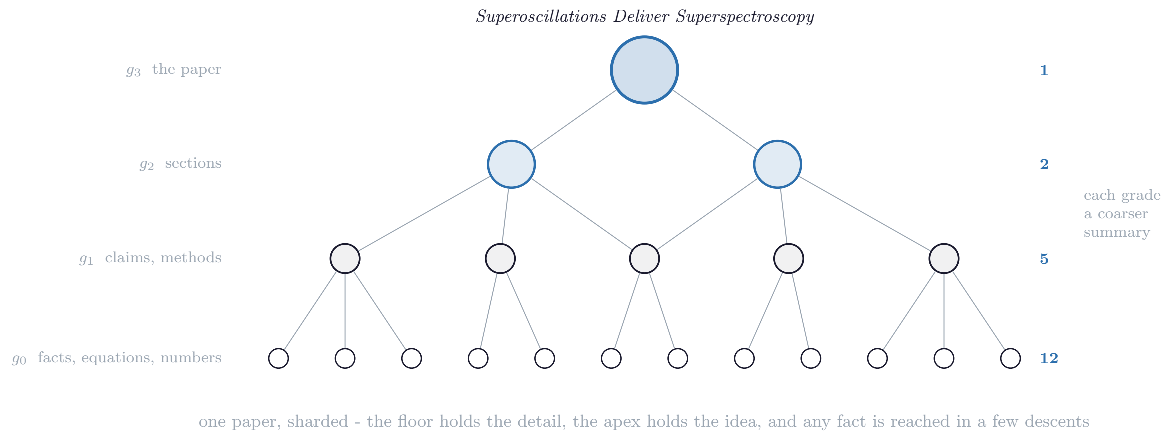


**Figure 2:** One paper, sharded. The floor holds the detail and the apex holds the idea; the levels shrink at each step, which is what makes the depth logarithmic and the recall cheap.

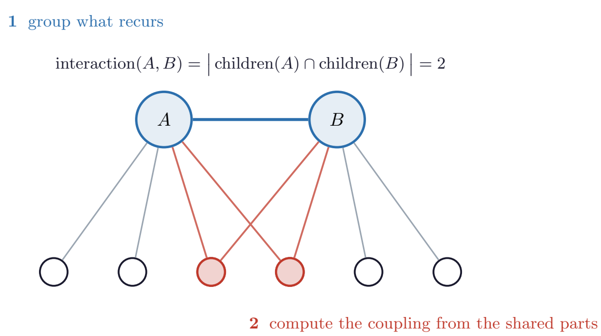


**Figure 3:** Coupling from shared children. Two theme-summary nodes $A$ and $B$ are linked whenever their child sets overlap; the coupling strength is $|\mathrm{children}(A) \cap \mathrm{children}(B)|$. The red nodes are the two shared children; the blue edge records the link. One grouping step yields the coupling matrix; no model is called.

one child ($C_2$), so their coupling is 1; a cross-author query resolves to a set intersection. Coupling between two theme-summary nodes $A$ and $B$ is the count of children they share: $\mathrm{interaction}(A, B) = \left|\mathrm{children}(A) \cap \mathrm{children}(B)\right|$ (Figure 3). In the corpus reported here, 69 theme-summary nodes cover 1,307 papers (papers counted once per corpus they appear in, so a paper shared across two corpora contributes two instances), a ratio of 5.3 percent, and a cross-author thematic link query executes zero model inferences. Of the 18 cross-researcher links recovered this way, 11 are citation-free, connecting researchers who have never cited one another and so are invisible to any bibliographic method. A fact that fits an existing level is absorbed in one step, as a schema in place lets a new memory settle in a day where an unfamiliar one would take weeks [29].

## 5 Generation, and the limits

Generating an artefact reverses the descent (Figure 4). A request enters at the top level, is matched against theme-summary nodes, and breaks into sub-questions for the level below, which identify the atomic propositions that must be read; the descent stops as soon as the available structure is enough to compose a reply, and the answers return enriched. A question about two researchers' shared methods arrives at the apex, where set lookup returns the two matching theme nodes. Each names the clusters beneath it; only those clusters are read and the answer assembled from their summaries. The generating direction mirrors how the brain processes vision: predictions pour down a hierarchy and only the prediction errors return [30, 31]. Because the structure was built from one person's own documents, what returns already carries the conceptual groupings their work has established; that structure shapes what returns before any prose is composed. Prior approaches have added explicit memory for agents as paged stores and remembered streams of experience [32, 33]; memory built as a graded, reconstructive structure is what lets a system's generation feed on its own past. The 11 citation-free links were unknown before the query ran.

The limits are structural. Building the graded representation and querying it cheaply (persistence) is solved. Voice and reasoning remain open: the prose is still generated by a general-purpose model whose weights reflect its training corpus, and closing that gap means training an open-weight model on the stored structure, which has not been done. Larger corpora, finer grading, and a self-hosted generator all require compute beyond what this paper demonstrates. Figure 5 shows the shape the closed system takes: a small local model walks the question down the structure and spends one call on a frontier model accessed at per-call cost (the large,

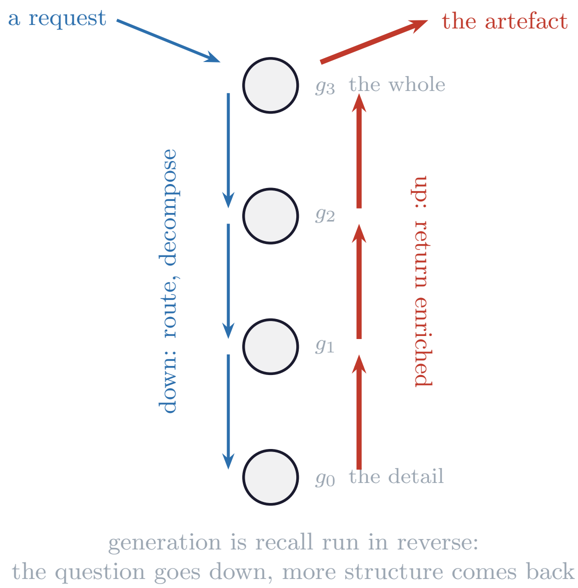


**Figure 4:** The grade ladder for generation. The question descends through the grades; the answers return enriched. To generate is to run recall in reverse.

externally-hosted model whose compute is rented by the call – hereafter rented intelligence), handing it a brief the structure has already half-composed, so that rented intelligence never sees the corpus and is paid by the sentence. The retrieval half is demonstrated; the generation half remains open.

## 6 Measured results

Every number is produced by `corpus_experiments.py` and every figure by `corpus_figures.py`. Authors are shown by anonymised identifier, A the author and 1–7 the collaborators, with the key in the appendix.

Counting every node by its grade gives 4,829 atomic propositions at the floor, then 637, 144, 23 as the fold rises. The successive ratios are 7.6, 4.4, and 6.3; their geometric mean gives a mean branching factor of $b \approx 5.7$ as estimated from adjacent levels, or $b \approx 4.75$ from a fit to the full level counts. Either way the depth to reach any of the 5,662 nodes is near $\log_b N \approx 5.5$ (Figure 6). The few nodes at the highest grades summarise relationships across multiple document corpora; they are built in a second pass that links theme-summary nodes across corpora when their child sets overlap, and are flagged in the script output and drawn in grey.

Across eight corpora built independently, from 29 to 388 papers, the number of theme-summary nodes grows far slower than the number of papers (Figure 7). An indicative power-law fit over eight points gives an exponent near 0.23 – an exponent of 0.23 means doubling the corpus grows the apex by only about 17% ($2^{0.23} \approx 1.17$), far less than doubling. The fit explains just over half the variance across the eight corpora ($R^2 \approx 0.56$); a larger set of corpora is needed to pin the exponent. Larger corpora are compressed harder: the exponent below one means the apex grows slower than the corpus.

Taking each of the eight authors in turn as the query centre, the fraction of recovered cross-links that are citation-free ranges from 0.61 to 1.00, mean 0.875 (Figure 8). The lowest belongs to the author, consistent with the author having the most co-authorship overlap with the collaborators in this corpus: the same mechanism that finds citation-free links also recovers real co-authorships where they exist. The eight-centre range (0.61–1.00) shows the result holds across the corpus.

## 7 How this paper was generated

Two conceptrics were held before a word was drafted. The *form conceptric* encoded the manuscript's structural demands: section shapes, figure standards, caption conventions, the house template, and the projection rules that govern how each section must sit relative to the others. The *content conceptric* carried the measured results – every number linked by name to `corpus_experiments.py`, every figure to `corpus_figures.py` – closed under reproduction: any number regenerates from its named script without human intervention. Literature was materialised into the content conceptric one reference at a time, each claim held against an external source until no assertion was unsupported.

The projector then folded form against content to draft each section. The draft entered a panel of four sealed judges (each judge is a separate model call with a fixed, unedited system prompt – the seal – returning a binary pass/flag with the flagged span quoted): one for voice (British register; no house-style defaults), one for closure (every term grounded before use, no specialist assumed), one for caption-truth (each caption checkable against the figure it names), one for naked claims (no assertion without a number or a reference). Where a judge flagged, the flag became the specification for a rewrite of that section alone; the projector recurred on the revised content until the panel returned silence. The author supplied the method and designed the judges. He did not edit the prose between iterations (Figure 9).

Every number in the paper reproduces from a named script; every figure from another. The method that assembles the brief, the data-ownership arrangement under which the personal corpus is held privately, and the open-weight model that would generate the prose without rented compute are the subject of ongoing work. The artefact and its reproducibility are what is delivered here.

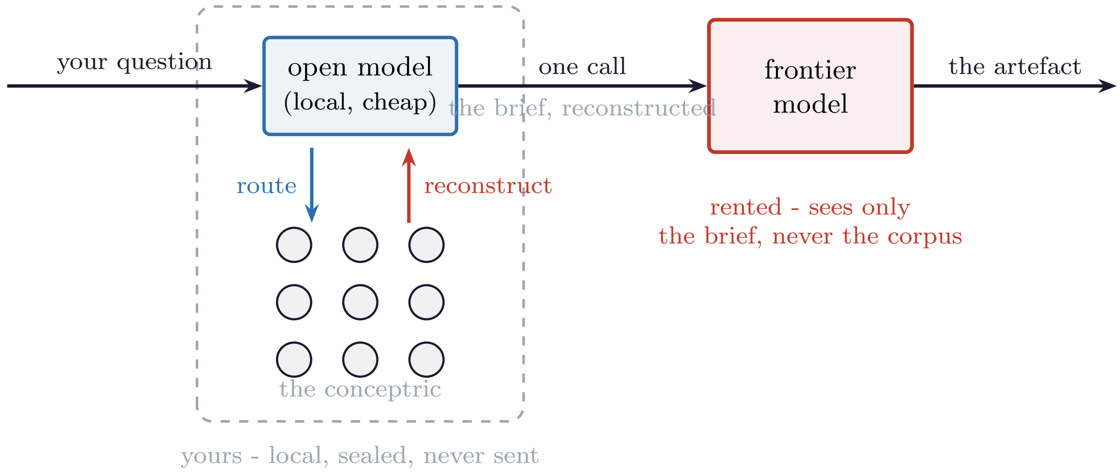


**Figure 5:** The architecture running. A small open model routes the question down the structure and reconstructs the few things that bear on it, then spends one call on the frontier model. The frontier model never sees the corpus; it answers a brief assembled by a machine that knows you.

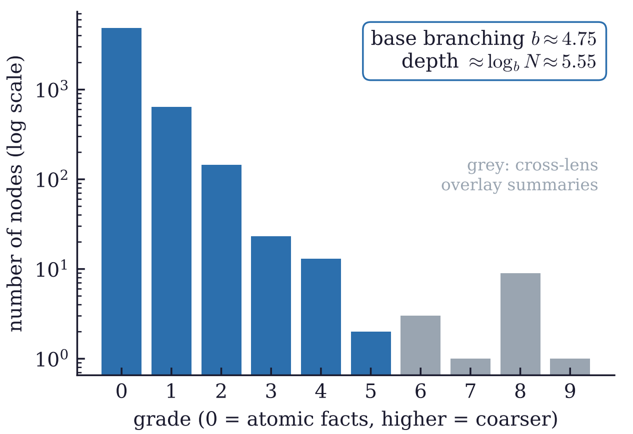


**Figure 6:** The grade histogram (log scale). The base ladder (blue) falls geometrically with mean branching factor $b \approx 4.75$, putting the depth to reach any node near $\log_b N \approx 5.5$. Grey bars are cross-corpus overlay summaries.

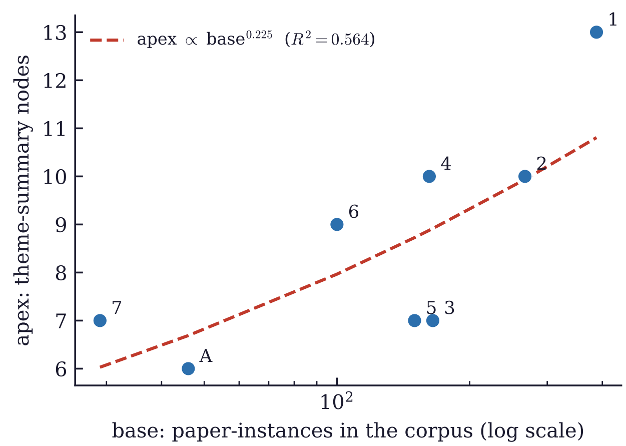


**Figure 7:** Theme-summary (apex) count against paper count (base) across eight independently built corpora, by author identifier. Apex size grows sublinearly with corpus size.

## What this gains over a retrieval-augmented or harness approach

This system is a recursive RAG: it folds retrieved material into a persistent hierarchy [26, 27], rather than querying a flat index on each call.

A standard retrieval-augmented generation setup [25] holds its documents in a flat, stateless index. At query time it embeds the query, runs a nearest-neighbour search over the index to surface passages semantically close to the question, assembles those passages into a prompt, and spends a model call to reason over them. Each query costs one embedding and one inference; the index carries no persistent structure across sessions; and what is retrieved is always an existing passage – a span of text that was already in a document. Cross-author thematic links of the kind reported in Section 6 appear in no single document and therefore cannot be retrieved by any method that retrieves spans of existing text, since no single document contains the link.

RAPTOR [26] builds a summarisation tree over clusters of passages – the closest prior art. The difference is persistence: RAPTOR rebuilds the tree per corpus and does not accumulate structure across sessions, so returning sessions re-pay the clustering cost and cross-session structural queries are unavailable. A harness of orchestrated agents [32] persists memory across ses-

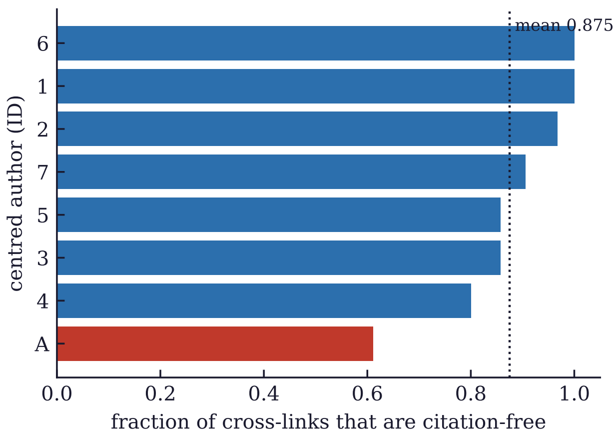


**Figure 8:** Fraction of cross-links that are citation-free, by centred author; the dotted line is the mean. The author (A) is lowest because real co-authorships are correctly recovered there.

sions via an OS-style paging mechanism, but the store remains a flat retrieval index without graded hierarchical structure; structural relations between documents are not first-class objects and cannot be recovered because the graph was never built.

The substrate here persists: the graded graph accumulates across sessions, structural relations are first-class objects, and the 11 citation-free cross-author links recovered are properties of the structure – they appear in no document, and retrieval of existing text therefore misses them by construction.

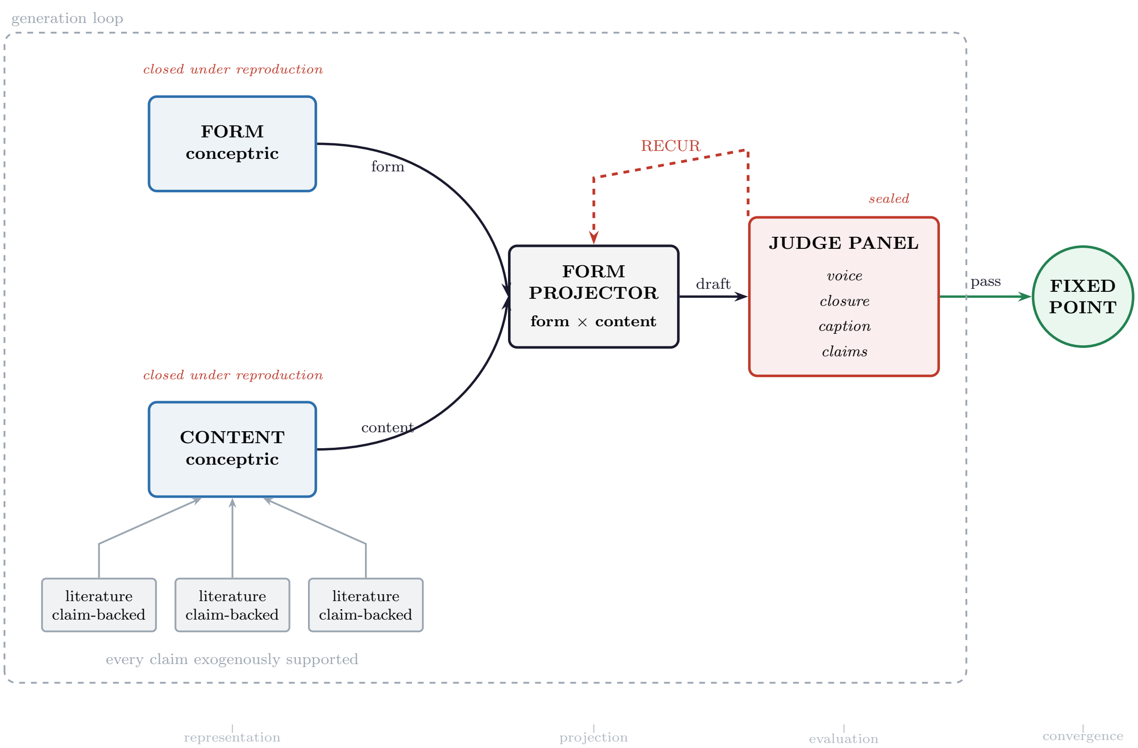


**Figure 9:** The projection process. Two conceptrics are held separately: a form conceptric encoding the manuscript's structural demands (section shapes, figure standards, the house template) and a content conceptric carrying the measured results with each number linked to its generating script. The projector folds them together to draft each section. A panel of sealed judges – voice, closure, caption-truth, naked-claims – scores the output; the loop recurs until the panel is silent.

# A The corpus and its cross-links

The eight corpora were assembled independently from public records. The author is labelled A; the seven collaborators are numbered by corpus size. Table 1 gives the key; the body and the data figures use only the identifiers.

| ID | Author | Themes | Papers |
|---|---|---|---|
| A | Gerard McCaul *(the author)* | 6 | 46 |
| 1 | Tapio Ala-Nissila | 13 | 388 |
| 2 | Kurt Jacobs | 10 | 267 |
| 3 | Alexandre Balanov | 7 | 165 |
| 4 | Denys Bondar | 10 | 162 |
| 5 | Juan Sebastian Totero Gongora | 7 | 150 |
| 6 | George H. Booth | 9 | 100 |
| 7 | Alicia Magann | 7 | 29 |

**Table 1:** The corpus: the author (A) and seven collaborators, numbered by corpus size. Names appear only in this appendix.

### Research themes by author.

**A** Quantum-classical unification: a single dynamical framework; Open quantum systems: exact dynamics from equilibrated initial states; Laser-driven control of correlated matter: prescriptive and inverse approaches; Superoscillations as a spectroscopic and confinement resource; Quantum simulation and computation: efficient algorithms for spectra and dynamics; Reservoir and machine learning with quantum/photonic substrates.

**1** Surface diffusion, adsorbate ordering & surface phase transitions; Heteroepitaxial thin-film growth, dislocations & surface morphology; Polymer translocation through nanopores: dynamics, scaling & sequencing; Ionic transport, polyelectrolyte electrostatics & electrokinetics beyond mean field; Phase-field crystal models, mesoscale crystallography & grain-boundary physics; Machine-learned interatomic potentials: neuroevolution, GAP & universal models; Thermal transport: phonon physics, MD methods, nanofluids & heat engineering; Interface roughening, kinetic growth & scaling universality (KPZ and beyond); Soft matter, colloids, hydrodynamics & complex fluids; Quantum thermodynamics, open quantum systems & quantum information devices; Nanoparticle & nanoscale optics: plasmonics, Mie scattering & radiative transport; Epidemic dynamics, mobility modelling & complex systems; Pseudorandom number generator testing & Monte Carlo reliability.

**2** Quantum Measurement as Control: Continuous Measurement, State Estimation, and Feedback; Quantum-to-Classical Transition: Decoherence, Measurement, and the Emergence of Classical Dynamics; Quantum Control Theory: Optimal, Coherent, and Time-Optimal Protocols; Quantum Thermodynamics and Information: Measurement Energetics, Feedback Work, and Maxwell's Demon; Quantum State Engineering in Mesoscopic and Mechanical Systems; Quantum Networking and Photonic Quantum Information: Gates, Transduction, and Communication; Quantum Error Correction, Decoherence-Free Subspaces, and Noise Robustness; Quantum Metrology and Nonclassicality as a Resource; Quantum Information Theory: Entanglement, Information Bounds, and Foundations; Stochastic Methods, Open Systems Pedagogy, and Mathematical Foundations.

**3** Nonlinear dynamics and chaos in semiconductor superlattices; Synchronization phenomena, phase dynamics, and bifurcations in coupled oscillators; Noise-driven dynamics, stochastic resonance, and delayed-feedback control; Neuromorphic hardware: memristive neurons, spiking dynamics, and neural computation; Quantum systems: chaos, hyperchaos, transport, and quantum sensing/computing; Physical reservoir computing and neuromorphic device platforms; 2D materials, nanostructures, and novel device physics.

**4** Unified Quantum-Classical Framework (Operational Dynamic Modeling); Quantum Control and Inverse Design of Dynamics; Open Quantum Systems, Dissipation Engineering, and Master Equations; Phase-Space Representations and the Wigner / Husimi Programme; Strong-Field Physics and Laser-Driven Ionization; High Harmonic Generation as a Many-Body Spectroscopic and Computational Probe; Superoscillations and Sub-Bandwidth Sensing / Imaging; Relativistic Quantum Mechanics and Quantum Electrodynamics Extensions; Two-Center Coulomb Problem and Dimensional Analysis; Computational Methods for Quantum Many-Body and Numerical Algorithms.

**5** Field-Sensitive THz Imaging and Spatiotemporal Control; Nonlinear THz Generation: Phase-Matching-Free and Surface-Enhanced Mechanisms; Laser Cavity-Soliton Microcombs: Self-Starting, Stability, and Metrology; Nanolasers and Radiationless Electromagnetic States (Anapoles, BSC, Spasers); Disorder as Design Principle in Nanophotonic and Plasmonic Systems; Photonic Reservoir Computing and Computational Photonics; Nonlinear Wave Dynamics: Collapse, Solitons, and Stability Theory.

| pair | sim | theme (left) | theme (right) | link |
|---|---|---|---|---|
| A–5 | 0.25 | Reservoir and machine learning with quantum/photonic substrates | Photonic Reservoir Computing and Computational Photonics | shared paper |
| A–4 | 0.22 | Superoscillations as a spectroscopic and confinement resource | Superoscillations and Sub-Bandwidth Sensing / Imaging | shared paper |
| A–6 | 0.20 | Laser-driven control of correlated matter: prescriptive and inverse approaches | Nonequilibrium and driven correlated systems: attosecond dynamics, high-harmonic generation, and quantum control | citation-free |
| A–7 | 0.20 | Quantum simulation and computation: efficient algorithms for spectra and dynamics | Quantum many-body spectral property estimation via imaginary-time methods | citation-free |
| 3–5 | 0.17 | Nonlinear dynamics and chaos in semiconductor superlattices | Nonlinear THz Generation: Phase-Matching-Free and Surface-Enhanced Mechanisms | citation-free |
| 1–3 | 0.15 | Quantum thermodynamics, open quantum systems & quantum information devices | Quantum systems: chaos, hyperchaos, transport, and quantum sensing/computing | citation-free |
| 1–2 | 0.15 | Thermal transport: phonon physics, MD methods, nanofluids & heat engineering | Quantum State Engineering in Mesoscopic and Mechanical Systems | citation-free |
| A–3 | 0.14 | Reservoir and machine learning with quantum/photonic substrates | Physical reservoir computing and neuromorphic device platforms | shared paper |
| 2–7 | 0.12 | Quantum Control Theory: Optimal, Coherent, and Time-Optimal Protocols | Hybrid quantum-classical algorithms for optimal control and simulation | citation-free |
| 6–7 | 0.12 | General-purpose electronic structure platforms and benchmarking infrastructure | Standardized benchmarking infrastructure for quantum algorithms and hardware | citation-free |
| 5–7 | 0.12 | Disorder as Design Principle in Nanophotonic and Plasmonic Systems | Hybrid quantum-classical algorithms for optimal control and simulation | citation-free |
| 1–6 | 0.12 | Quantum thermodynamics, open quantum systems & quantum information devices | Correlated electronic structure on quantum hardware: VQE, quantum embedding, and near-term algorithms | citation-free |

**Table 2:** The strongest thematic cross-link between each author pair (top twelve of 28). Most connect researchers who share no paper.

**6** Exact stochastic many-body methods: pushing FCI-quality electronic structure to larger systems; Quantum embedding: systematically improvable multiscale treatments of correlated fragments in extended environments; Spectral and dynamical properties from static methods: Green's functions, self-energies, and moment-conserving spectra; Machine-learning-inspired and tensor-network ansätze: compact, expressive wave function representations; Correlated electronic structure on quantum hardware: VQE, quantum embedding, and near-term algorithms; Nonequilibrium and driven correlated systems: attosecond dynamics, high-harmonic generation, and quantum control; Wave function interpolation and machine-learning potentials: correlated PES and nonadiabatic dynamics at scale; General-purpose electronic structure platforms and benchmarking infrastructure; Out-of-scope / non-research-program papers.

**7** Feedback-based quantum control without classical optimization; Quantum tracking control for molecular systems; Hybrid quantum-classical algorithms for optimal control and simulation; Variational quantum algorithm landscape theory; Standardized benchmarking infrastructure for quantum algorithms and hardware; Control engineering applied to behavioral health and mobile health interventions; Quantum many-body spectral property estimation via imaginary-time methods.

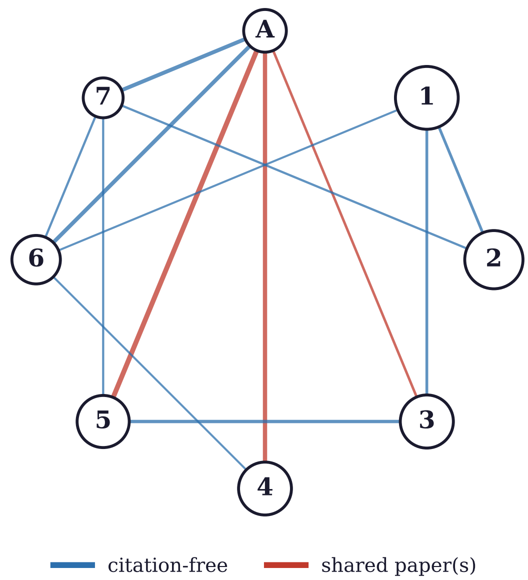


**Figure 10:** The cross-link network. Each node is an author (A the author, 1–7 collaborators); each edge is the strongest thematic link between a pair, drawn only above a normalised similarity of 0.10 (raw child-overlap counts normalised by the geometric mean of each node's child count, giving a Jaccard-like score in $[0, 1]$). Blue edges are citation-free; vermilion edges connect authors who share a paper.

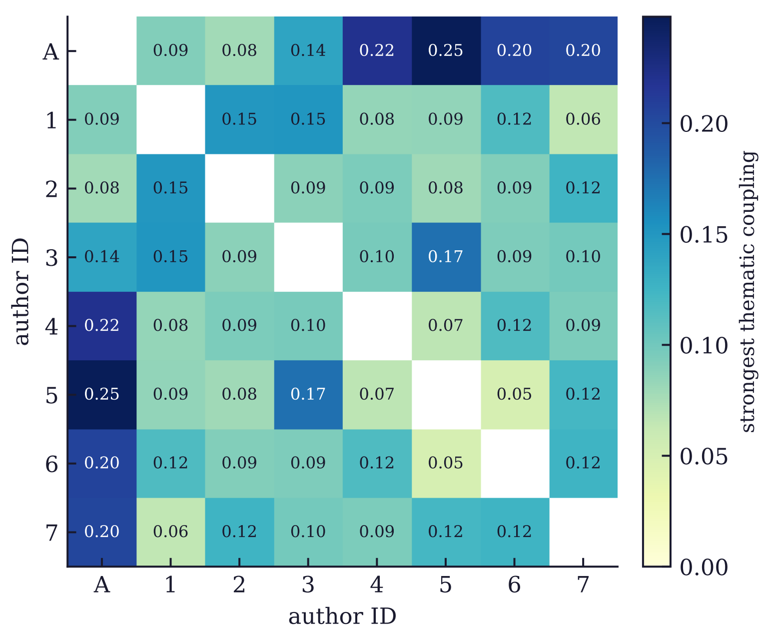


**Figure 11:** Strongest thematic coupling between every pair of authors. The author (A) couples most strongly to collaborators 5 and 4 (0.25 and 0.22), with 6 and 7 close behind; the self-diagonal is omitted.

*This document was generated by **EigenEngine**. Gerard McCaul wrote none of its words.*

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